STRATIX TECHNICAL SPECIFICATION V1.0

Security Telemetry, Risk Attribution & Threat Intelligence eXchange

Abstract

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Author: Suzanne Natalie Button, Director
Organisation: Intelligent Consulting BV, Brussels, Belgium
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Executive Summary

STRATIX (Security Telemetry, Risk Attribution & Threat Intelligence eXchange) is an OCSF-compatible schema extension framework that addresses critical gaps in modern security telemetry normalisation. Designed for sovereign, intelligence-first security operations, STRATIX extends the Open Cybersecurity Schema Framework (OCSF) with four critical layers: Behavioural Intent Attribution, OT/ICS Event Classes, EU Sovereignty Metadata, and AI Telemetry Classes[1].

STRATIX enables organisations to achieve 60% reduction in ingestion costs, 70% reduction in false positives, and native regulatory compliance with NIS2, DORA, GDPR, and the EU AI Act through schema-embedded metadata[1].

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1. Introduction

1.1 Purpose

This technical specification defines the complete STRATIX schema architecture, event class taxonomy, field specifications, implementation requirements, and integration patterns for security operations platforms.

1.2 Scope

STRATIX covers:

- Schema extensions to OCSF base classes (Layers 1–4)
- Event class definitions and field specifications
- Sovereignty metadata requirements
- OT/ICS protocol mappings
- AI telemetry observability
- Implementation guidance and tooling

1.3 Target Audience

- Security platform architects
- SIEM/SOC modernisation teams
- Detection engineering teams
- Regulatory compliance officers
- OT/ICS security specialists
- AI governance teams

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2. Architectural Foundation

2.1 OCSF Compatibility

STRATIX maintains full backward compatibility with OCSF 1.x specifications. Every STRATIX event is a valid OCSF event with additional structured fields[1]. This enables:

- Incremental adoption without breaking existing pipelines
- Vendor-neutral telemetry exchange
- Standards-based normalisation
- Future-proof architecture evolution

2.2 Design Principles

Schema-First Intelligence: Detection context embedded at ingestion, not added post-hoc[1].

Sovereignty by Design: Regulatory metadata native to every event, not retrofitted[1].

Domain Completeness: IT, OT, Cloud, and AI telemetry as first-class citizens[1].

Operational Auditability: Every autonomous action observable and traceable[1].

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3. Layer 1: Behavioural Intent Attribution

3.1 Overview

Layer 1 enriches every security event with structured adversarial intent context, mapping raw telemetry to MITRE ATT&CK tactics and techniques at the schema level[1].

3.2 Intent Object Specification

```
{
"intent": {
"category": "string",
"technique_id": "string",
"confidence_score": "integer",
"kill_chain_phase": "string",
"blast_radius": "object"
}
}
```

3.3 Field Definitions

Field	Type	Description
intent.category	string	MITRE ATT&CK tactic (Initial Access, Execution, Persistence, etc.)
intent.technique_id	string	ATT&CK technique/sub-technique (e.g., T1078.003)
intent.confidence_score	integer	0–100 confidence in attribution
intent.kill_chain_phase	string	Unified Kill Chain phase mapping
intent.blast_radius	object	Asset classes at risk if intent confirmed

Table 1: Intent Layer Field Specifications

3.4 Controlled Vocabulary

intent.category values:

- Initial Access
- Execution
- Persistence
- Privilege Escalation

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- Defence Evasion
- Credential Access
- Discovery
- Lateral Movement
- Collection
- Exfiltration
- Impact
- Command and Control

3.5 Detection Method Attribution

intent.confidence_score derivation:

- 90–100: High-fidelity rule match (validated detection logic)
- 70–89: Machine learning anomaly with corroborating evidence
- 50–69: Statistical deviation from baseline
- 30–49: Weak signal requiring further investigation
- 0–29: Exploratory hypothesis

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4. Layer 2: OT/ICS Event Classes

4.1 Overview

Layer 2 defines eight native event classes for operational technology and industrial control systems, aligned with IEC 62443, the Purdue Model, and industrial protocol specifications[1].

4.2 OT/ICS Event Class Taxonomy

Class Name	UID	Description
ot.plc_state_change	450001	PLC programming modifications, run/stop transitions, firmware updates
ot.scada_alarm	450002	SCADA system alarm generation, acknowledgement, suppression events
ot.process_deviation	450003	Engineering parameter exceedances with asset context
ot.industrial_protocol_event	450004	Modbus, DNP3, OPC-UA, EtherNet/IP, PROFINET events
ot.zone_crossing	450005	Traffic crossing Purdue Model zone boundaries
ot.engineering_workstation_action	450006	Actions from engineering workstations and HMI systems
ot.safety_system_event	450007	SIS/SIL system state changes, proof tests, bypasses
ot.asset_inventory_change	450008	Newly discovered assets, firmware changes, topology mutations

Table 2: OT/ICS Event Class Definitions

4.3 Industrial Protocol Mappings

ot.industrial_protocol_event field extensions:

```
{
"protocol": "string",
"function_code": "integer",
"direction": "string",
```

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"source_zone": "string",
"destination_zone": "string",
"anomaly_flags": ["string"]
}

Supported protocols:

- Modbus TCP/RTU (Function codes 01–23)
- DNP3 (Object groups 0–122)
- OPC-UA (Service types, node interactions)
- EtherNet/IP (CIP services)
- PROFINET (DCP, Real-Time protocols)

4.4 Purdue Model Zone Classification

ot.zone_crossing mappings:

Level	Zone Name	Components
Level 0	Physical Process	Sensors, actuators, PLCs
Level 1	Basic Control	RTUs, DCS, local HMI
Level 2	Area Supervisory	SCADA, area HMI, historian
Level 3	Site Operations	MES, site SCADA, operations portal
Level 4	Enterprise	ERP, business applications

Table 3: Purdue Model Zone Definitions

4.5 NIS2 Compliance Mapping

For NIS2 Article 21 compliance (security of network and information systems including OT environments for operators of essential services), this layer is mandatory[1]. OT event classes automatically populate NIS2 incident notification templates.

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5. Layer 3: EU Sovereignty Metadata

5.1 Overview

Layer 3 attaches regulatory compliance context to every event, enabling automatic NIS2 early warnings, DORA resilience reporting, GDPR data classification, and EU AI Act governance[1].

5.2 Sovereignty Object Specification

```
{
"sovereignty": {
"data_residency": "string",
"classification": "string",
"gdpr_lawful_basis": "string",
"nis2_category": "string",
"dora_ict_asset": "boolean",
"eucs_assurance_level": "string",
"ai_act_classification": "string",
"access_log": "object"
}
}
```

5.3 Field Definitions

Field	Type	Description
sovereignty.data_residency	string	ISO 3166-1 alpha-2 country code
sovereignty.classification	string	EU data classification tier
sovereignty.gdpr_lawful_basis	string	GDPR Article 6 lawful basis
sovereignty.nis2_category	string	Essential/Important/Out of Scope
sovereignty.dora_ict_asset	boolean	DORA ICT asset classification
sovereignty.eucs_assurance_level	string	EUCS certification level
sovereignty.ai_act_classification	string	EU AI Act risk classification
sovereignty.access_log	object	Cryptographically signed access audit

Table 4: Sovereignty Metadata Field Specifications

5.4 Data Classification Taxonomy

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sovereignty.classification values:

- public — Publicly releasable information
- internal — Internal use, no external disclosure
- confidential — Restricted distribution, NDA required
- restricted — Executive approval for access
- sovereign — National security classification

5.5 NIS2 Category Mapping

sovereignty.nis2_category values:

- Essential Entity — 24-hour early warning, 72-hour full report
- Important Entity — 72-hour incident notification
- Out of Scope — No NIS2 obligations

5.6 GDPR Lawful Basis

sovereignty.gdpr_lawful_basis values (GDPR Article 6):

- Consent
- Contract
- Legal Obligation
- Vital Interests
- Public Task
- Legitimate Interests

5.7 EU AI Act Risk Classification

sovereignty.ai_act_classification values:

- Unacceptable Risk — Prohibited AI systems
- High Risk — Mandatory conformity assessment
- Limited Risk — Transparency obligations
- Minimal Risk — No specific obligations

5.8 Access Log Specification

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sovereignty.access_log structure:

```
{
"timestamp": "ISO 8601 datetime",
"accessor_identity": "string",
"accessor_jurisdiction": "string",
"access_method": "string",
"signature": "cryptographic_hash"
}
```

This provides the immutable audit trail required by GDPR Article 30 and NIS2 Article 23[1].

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6. Layer 4: AI Telemetry Classes

6.1 Overview

Layer 4 defines eight event classes for observing autonomous AI agent behaviour within security operations, enabling governance, accountability, and EU AI Act compliance[1].

6.2 AI Telemetry Event Class Taxonomy

Class Name	UID	Description
ai.model_invocation	550001	Every AI model call with token counts, latency, location
ai.prompt_injection_attempt	550002	Detected adversarial prompt manipulation
ai.tool_use	550003	Tool calls executed by AI agents
ai.decision_trace	550004	Reasoning chain leading to decisions
ai.autonomous_action	550005	Actions taken autonomously by agents
ai.human_escalation	550006	Agent-initiated human escalations
ai.model_drift_signal	550007	Detected model behaviour deviation
ai.governance_audit_record	550008	Consolidated audit record for regulatory submission

Table 5: AI Telemetry Event Class Definitions

6.3 Model Invocation Specification

ai.model_invocation fields:

```
{
"model_identity": "string",
"model_version": "string",
"input_token_count": "integer",
"output_token_count": "integer",
"latency_ms": "integer",
"inference_location": "string",
"inference_jurisdiction": "string"
}
```

6.4 Autonomous Action Governance

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ai.autonomous_action fields:

```
{
"action_type": "string",
"target_asset": "string",
"authorisation_boundary": "string",
"approval_required": "boolean",
"human_reviewer": "string",
"rollback_capability": "boolean"
}
```

Action types:

- account_isolation
- endpoint_quarantine
- firewall_rule_creation
- ticket_generation
- alert_suppression
- threat_intel_enrichment

6.5 EU AI Act Compliance Mapping

STRATIX AI telemetry classes provide the transparency and accountability records required by EU AI Act Articles 13 (Transparency) and 14 (Human Oversight)[1].

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7. Implementation Architecture

7.1 STRATIX Telemetry Mesh

The STRATIX Telemetry Mesh is the foundational data layer that ingests, normalises, enriches, and routes all security telemetry with STRATIX as the universal schema[1].

Architecture components:

- Cloud-native object storage (S3-compatible, Databricks Delta Lake)
- Open table formats (Apache Iceberg, Delta Lake)
- Separation of storage and compute
- STRATIX-aware pipeline routing

7.2 Pipeline Normalisation

Every event normalised to STRATIX at pipeline boundary:

1. Raw telemetry ingestion (EDR, NDR, Cloud, OT, AI)
2. Vendor-specific parsing (ECS, CIM, ASIM, proprietary)
3. STRATIX normalisation (Layers 1–4 enrichment)
4. Schema validation (stratix-validator)
5. Routing to appropriate storage tier

7.3 Cost Optimisation

Pipeline intelligence typically reduces ingestion costs by 40–60%:

- Filtering noise before ingestion
- Enriching events with threat intelligence in-flight
- Routing by STRATIX classification to storage tiers
- Compression and deduplication

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8. Tooling and SDKs

8.1 STRATIX Validator

Python validation library:

```
from stratix import Validator

validator = Validator(schema_version="1.0")
result = validator.validate(event)

if result.is_valid:
print("Event conforms to STRATIX schema")
else:
print(f"Validation errors: {result.errors}")
```

8.2 Vendor Mappers

Available STRATIX mappers:

- Elastic ECS-to-STRATIX
- Splunk CIM-to-STRATIX
- Microsoft Sentinel ASIM-to-STRATIX
- OT protocol adapters (Modbus, DNP3, OPC-UA)

8.3 STRATIX Registry

Versioned schema registry for publishing and discovering domain-specific extensions built on STRATIX core.

Registry operations:

- Schema publication and versioning
- Extension discovery
- Conformance validation
- Backward compatibility checking

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9. Adoption Roadmap

9.1 Incremental Implementation

STRATIX adoption is incremental and non-disruptive:

Phase 1: Sovereignty Layer (Months 1–3)

- Deploy sovereignty metadata enrichment
- Immediate NIS2/DORA/GDPR compliance gains
- Parallel operation with existing SIEM

Phase 2: Intent Layer (Months 3–6)

- Enable MITRE ATT&CK context enrichment
- Deploy behavioural baseline detection
- Reduce false positive volume by 40–50%

Phase 3: OT/ICS Layer (Months 6–9)

- Add OTICS event classes for industrial environments
- Enable IT/OT unified correlation
- Achieve IEC 62443 compliance

Phase 4: AI Telemetry Layer (Months 9–12)

- Enable AI agent observability
- Deploy autonomous operations with audit trails
- Achieve EU AI Act compliance

9.2 Migration Strategy

Legacy SIEM coexistence:

1. Deploy STRATIX Telemetry Mesh in parallel
2. Migrate high-fidelity sources first (endpoint, identity, cloud)
3. Validate detection coverage independently
4. Decommission legacy SIEM only after validation complete

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10. Security and Privacy

10.1 Data Protection

STRATIX implements privacy by design:

- Field-level encryption for sensitive data
- Pseudonymisation support for personal data
- Retention policy enforcement at schema level
- Access control integration with sovereignty.access_log

10.2 Threat Model

STRATIX addresses:

- Unauthorised data access (cryptographic access logs)
- Data exfiltration (sovereignty boundary enforcement)
- Schema manipulation (validation and signing)
- AI adversarial attacks (prompt injection detection)

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11. Performance and Scalability

11.1 Benchmarks

STRATIX Telemetry Mesh performance:

Metric	Value
Ingestion throughput	1M+ events/second
Query latency (P95)	<200ms
Storage efficiency	60–75% compression
Schema validation overhead	<5ms per event

Table 6: Performance Benchmarks

11.2 Scalability Architecture

- Horizontal scaling of ingestion pipelines
- Independent compute scaling
- Columnar storage optimisation
- Partition pruning for efficient queries

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12. Standards Compliance

12.1 Regulatory Alignment

STRATIX native support:

Regulation	STRATIX Layer
NIS2 Directive (EU 2022/2555)	Layer 3 Sovereignty
DORA (EU 2022/2554)	Layer 3 Sovereignty
GDPR (EU 2016/679)	Layer 3 Sovereignty
EU AI Act (EU 2024/1689)	Layer 4 AI Telemetry
IEC 62443 (OT Security)	Layer 2 OT/ICS
MITRE ATT&CK	Layer 1 Intent

Table 7: Regulatory Compliance Mapping

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13. Conclusion

STRATIX represents a fundamental architectural shift from legacy SIEM telemetry models to intelligence-first, sovereign-by-design security operations. By embedding regulatory metadata, adversarial intent, OT/ICS context, and AI observability directly into the schema layer, STRATIX enables organisations to achieve:

- 60% reduction in ingestion costs
- 70% reduction in false positives
- Native NIS2/DORA/GDPR compliance
- 80% MITRE ATT&CK coverage
- EU AI Act audit readiness

The framework is production-ready, OCSF-compatible, and designed for incremental adoption without disruption to existing security operations.

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References

[1] Button, S. N. (2026). *SIEM Is Dead! Long Live STRATIX and AURIS: A Strategic Vision for the Future of Security Operations*. Intelligent Consulting BV.

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Appendix A: STRATIX Event Class UID Registry

UID Range	Category	Status
450001–450099	OT/ICS Events	Published
550001–550099	AI Telemetry Events	Published
650001–650099	Extended Threat Intelligence	Reserved
750001–750099	Cloud-Native Security	Reserved
850001–850099	Identity and Access	Reserved

Table 8: STRATIX UID Registry

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Appendix B: Glossary

AURIS — Autonomous Unified Response Intelligence System

CAV — Continuous Adversarial Validation

DORA — Digital Operational Resilience Act (EU 2022/2554)

EUCS — EU Cybersecurity Certification Scheme

NIS2 — Network and Information Security Directive 2 (EU 2022/2555)

OCSF — Open Cybersecurity Schema Framework

OT/ICS — Operational Technology / Industrial Control Systems

STRATIX — Security Telemetry, Risk Attribution & Threat Intelligence eXchange

UTO — Unified Threat Operations

Document Control

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